

Problem Set 4

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1) Introduction

The goal of this project is to test whether a firm-level characteristic called operating profitability (`op_at`) can predict future stock returns. The idea is simple: if we sort stocks into groups based on this measure and track how each group performs over the next month, we can see whether more profitable firms earn higher returns than less profitable ones. I also look at how this interacts with firm size (`market_equity`) using bivariate sorts.

2) Constructing the Portfolio

2.1 Return Calculation

Monthly returns are computed using adjusted closing prices over a 20-trading-day window. The adjustment uses the ratio of the current adjustment factor to its last known value to correct for corporate actions like stock splits. Excess returns subtract the 30-day Treasury yield to remove the compensation for simply holding a riskless asset.

2.2 Sorting Procedure

At each rebalancing date (the first available trading day of each month), stocks are ranked by the sorting variable and split into three groups using the 30th and 70th percentiles as breakpoints. Portfolio returns are market-cap weighted: each stock inside a group contributes proportionally to its market capitalization. This means larger firms have more influence on the portfolio return, which is common in index-oriented research. The long-short portfolio (`Diff`) goes long the highest tertile and short the lowest tertile. If `op_at` truly predicts returns, this spread should be positive and statistically significant.

2.3 Bivariate Sorts

In the bivariate dependent sort, stocks are first sorted into three groups by `op_at`, and then within each `op_at` group they are sorted again into three groups by `market_equity`. This gives a 3×3 grid of 9 portfolios where the `market_equity` breakpoints depend on which `op_at` group we are in. In the bivariate independent sort, both variables are sorted simultaneously using breakpoints from the full cross-section, not conditioned on each other. Stocks are then assigned to the 3×3 cell they fall into based on both sorts.

3. Tests

To check whether the average portfolio returns are different from zero, we used a t-test with Newey-West standard errors using 6 lags. The reason for this adjustment is that monthly returns can be correlated with themselves from one month to the next. A regular t-test does not account for this and would give t-statistics that are too large. The Newey-West correction gives a better estimate of statistical significance.

4. Results

a) Univariate Sort

Portfolio	Avg Return	t-stat	p-value
op_at_1 (low profitability)	1.12%	2.91	0.004
op_at_2 (medium)	1.20%	4.71	<0.001
op_at_3 (high profitability)	1.50%	5.99	<0.001
Diff (High – Low)	0.38%	1.31	0.190

The pattern is in the expected direction: more profitable stocks earn higher returns. Portfolio 3 averages 1.50% per month while Portfolio 1 averages 1.12%. However, the long-short spread of 0.38% per month is not statistically significant ($p = 0.19$). This means we cannot rule out that the difference is due to random variation in the sample.

b) Bivariate Dependent Sort

	Small firms	Mid firms	Large firms	Size Diff
Low op_at	1.61%	1.26%	1.25%	-0.73%
Mid op_at	1.64%	1.24%	1.17%	-0.06%
High op_at	1.74%	1.53%	1.55%	-0.26%
op_at Diff	-0.01%	+0.13%	+0.45%	+0.47%

The double-difference portfolio earns +0.47% per month ($t = 1.57$, $p = 0.12$). This is still not significant, but the direction is consistent. One interesting finding is that within each profitability group, smaller firms earn more than larger ones — showing a size premium even within S&P 500 stocks.

c) Bivariate Independent Sort

	Small firms	Mid firms	Large firms	Size Diff
Low op_at	1.58%	1.28%	1.26%	−0.56%
Mid op_at	1.57%	1.27%	1.21%	−0.10%
High op_at	1.60%	1.57%	1.61%	−0.33%
op_at Diff	+0.21%	+0.30%	+0.45%	+0.23%

The double-difference here is +0.23% per month. The independent sort gives a smaller spread than the dependent sort because the portfolio composition changes when breakpoints are not recalculated within each group. The general pattern is the same: profitability spreads are larger among larger firms.

5) Economic Reasoning and Conclusion

There are two common explanations for why highly profitable firms may earn higher returns. The first explanation is related to risk. One argument is that profitable firms might be exposed to certain risks that are not fully captured by standard asset pricing models, which means investors require higher expected returns as compensation.

The second explanation is based on investor behavior. Investors may not fully recognize how persistent a firm's profitability can be and therefore underestimate the value of highly profitable companies. As the market gradually adjusts its expectations, stock prices increase and generate positive returns.

To conclude;

This project implemented univariate and bivariate portfolio sorts using operating profitability (op_at) and market capitalization (market_equity) as sorting variables, over S&P 500 stocks from 2012 to 2021. The main findings are: highly profitable stocks earn more per month than low-profitability stocks. In bivariate sorts, the profitability effect is slightly stronger among larger firms. Smaller firms within the index consistently outperform larger ones across all profitability groups, consistent with a size premium. The dependent and independent bivariate sorts produce similar results, with the dependent sort showing a slightly larger spread. Overall, op_at points in the right direction as a return predictor, but the S&P 500 setting limits the strength of the result.